

JAYENDRA PAWAR

Software Engineer — Backend (Go), Kubernetes & Distributed Systems

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WORK EXPERIENCE

IBM

Software Engineer II (Platform/Backend)

Oct 2024 – Present

Bengaluru, Karnataka

DBaaS Kubernetes Controllers

- Architected and owned a highly available, **Go-based Kubernetes controller** for **750+ workers**, automating **worker lifecycle management** (provisioning, replacement & decommissioning) and **reducing provisioning latency by 30%**
- Orchestrated **safety-gated cluster automation** for **maintenance & node upgrades**, automating unhealthy-node replacement and unclaimed-node cleanup while **improving cluster utilization by 25%** and **reducing recurring on-call escalations**
- Designed a **zone-aware auto-scaling strategy** balancing resource utilization vs. availability across **multi-AZ clusters**, sustaining **99.9% uptime** while **eliminating over-provisioning**
- Developed an **integration-test framework** on **dedicated Kubernetes clusters**, validating **platform controllers** for worker orchestration, reconciliation, scaling, resilience & recovery across **12+ managed data services** as a **production-readiness gate**

Context-Based Restrictions (CBR)

- Driving the **end-to-end development** of **CBR Dedicated Instances**, covering **RESTful API design**, **event-driven network IP updates**, **technical specifications** and **infrastructure planning**; delivered SDK/CLI support for **enterprise customers**
- Engineered **Instance IP state synchronization** using **Kubernetes informers**, **rate-limited queues** and **batch processing**, enabling deduplication, idempotency, retries, and failure isolation to meet a **10s CBR zone-consistency SLA**
- Implemented **CBR policy enforcement** for **instance-level network policies**, replacing broad service-wide permissions with **fine-grained access control** across **CDP production deployments**

DevOps, CI/CD & Infrastructure

- Led **Gen3-to-Flex migration & Sandbox decommission**, cutting **infra costs ~14% (\$166K savings)** while mitigating outage risks
- Built a **Slack-based ChatOps** platform automating **30+ Kubernetes** and **managed-services** workflows, **reducing operational toil** and accelerating incident resolution
- Standardized **Deployable Architecture (IaC) modules**, enabling **40% faster feature rollouts** and simplifying **IAM configuration** across teams while strengthening system resilience
- Established **GitOps-based CI/CD pipelines** with **GitHub Actions**, pre-merge validation, security scans, and quality gates, enabling **zero-downtime deployments** across **on-premises Kubernetes clusters**

Freshworks

Software Engineer (Backend)

Jul 2022 – Sept 2024

Bengaluru, Karnataka

Payment Services & Integrations

- Engineered a **highly available microservice** for **Stripe integration** in **FreshPay**, enabling critical **payment workflows** such as bind flow, authorization, tokenization, payments, and refunds; serving **4.9 million users** in Europe
- Developed a **scalable payment notification service**, incorporating **API specifications**, **ERD designs**, and workflows using **gRPC** and **schedulers**, supporting **150K+ customers**
- Led the **migration of payment services** to **EKS (Kubernetes)**, achieving a **30% cost reduction** while enhancing **deployment efficiency** and **scalability** for millions of daily transactions
- Accelerated **MySQL performance**, cutting **query execution time by 60%** through **indexing**, **partitioning**, and **execution plan optimization**. Implemented **CDC-based caching** to synchronize dynamic and static data in real time, **reducing API response time by 40%**
- Deployed **canary releases** across **5 microservices**, minimizing deployment risk and **reducing rollback rate by 15%** while ensuring adherence to release policies, performance benchmarks, and SLOs

EDUCATION

National Institute of Technology, Jaipur

Bachelor of Technology in Mechanical Engineering; Minor in Computer Science

Jul 2018 – Jun 2022

Jaipur, Rajasthan

- Awarded **Aditya Birla Scholarship** — ranked among the **top 15 candidates across all NITs (2021)**

SKILLS

- Programming Languages:** Go, Python, Rust, C/C++
- Backend & Distributed Systems:** REST APIs, gRPC, WebSockets, Kafka, RabbitMQ, Event-Driven Architectures
- Cloud & Infrastructure:** Kubernetes, Docker, Linux, Helm, Terraform, GitHub Actions, Jenkins, GitOps
- Databases & Observability:** MySQL (Indexing, Partitioning), Redis, Prometheus, Grafana, Monitoring & Alerting
- Core Areas:** System Design, Microservices, Reliability Engineering, Performance Optimization